

Def. Sequence / List / Tuple

A sequence — also known as a list or tuple — is an ordered collection of objects, typically called components or entries. When the # of components in the collection is 2, 3, 4, ..., or n , the sequence is called an (ordered) pair, triple, quadruple or n -tuple, respectively.

— we write sequences, lists, tuples using angle brackets $\langle \dots \rangle$

Ex: $\langle 1, 2, 6 \rangle$ 3-tuple or triple

$\langle \text{Montana}, \text{Texas} \rangle$ 2-tuple or pair

$\langle 1, 2 \rangle \neq \langle 2, 1 \rangle$ | $\langle 0, 1, 2 \rangle \neq \langle 2, 0, 1 \rangle$

$\langle 1, \text{Texas}, \Delta \rangle$ ✓

Some people uses paranthesis to represent tuples

$\langle 1, 2 \rangle = (1, 2)$

$(1, 1) \leftarrow$ This is a possible pair

Def cartesian product of two sets

The cartesian product of two sets A and B is the set $A \times B = \{ \langle a, b \rangle : a \in A \wedge b \in B \}$ containing all ordered pairs where the first element comes from Set A and the second element comes from set B .

$$\langle 0, 2 \rangle \neq \langle 2, 0 \rangle$$

$$\langle 0, 2 \rangle = \langle 0, 2 \rangle$$

ex. $A = \{0, 1, 2\}$ $B = \{2, 3\}$

$$A \times B = \{ \langle 0, 2 \rangle, \langle 0, 3 \rangle, \langle 1, 2 \rangle, \langle 1, 3 \rangle, \langle 2, 2 \rangle, \langle 2, 3 \rangle \}$$

$$B \times A = \{ \langle 2, 0 \rangle, \langle 2, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 0 \rangle, \langle 3, 1 \rangle, \langle 3, 2 \rangle \}$$

Ex. $\mathbb{R} \times \mathbb{R} = 2D \text{ plane} / \text{cartesian plane}$

$$\langle 0, 0 \rangle \in \mathbb{R} \times \mathbb{R}$$

$$\langle 1.11234, 2.012345 \rangle \in \mathbb{R} \times \mathbb{R}$$

$$\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$$

$$\{\text{red, blue}\} \times \{1, 2, 3\} = \{\langle \text{red}, 1 \rangle, \langle \text{red}, 2 \rangle, \langle \text{red}, 3 \rangle, \langle \text{blue}, 1 \rangle, \langle \text{blue}, 2 \rangle, \langle \text{blue}, 3 \rangle\}$$

$$\langle 1, \text{red} \rangle \notin \{\text{red, blue}\} \times \{1, 2, 3\}$$

$$\langle 1, \text{red} \rangle \in \{1, 2, 3\} \times \{\text{red, blue}\}$$

$$\mathbb{R}^3 = 3D \text{ space}$$

$$\{0, 1\} \times \{2, 1\} = \{\langle 0, 2 \rangle, \langle 0, 1 \rangle, \langle 1, 2 \rangle, \langle 1, 1 \rangle\}$$

$$\{0, 1\} \times \{2, 1\} \times \{2, 0\} = \{\langle 0, 2, 2 \rangle, \langle 0, 2, 0 \rangle, \langle 0, 1, 2 \rangle, \langle 0, 1, 0 \rangle, \langle 1, 2, 2 \rangle, \langle 1, 2, 0 \rangle, \langle 1, 1, 2 \rangle, \langle 1, 1, 0 \rangle\}$$

They are fundamentally equivalent but not exactly the same

$$\underbrace{(\{0, 1\} \times \{2, 1\})}_{\text{set}} \times \underbrace{\{2, 0\}}_{\text{set}} = \{\langle \langle 0, 2 \rangle, 2 \rangle, \langle \langle 0, 2 \rangle, 1 \rangle, \dots \}$$

Question : What is the size of

$$|A \times B| = |A| \cdot |B|$$

8								
7								
6								
5				.				
4								
3								
2								
1								
	a	b	c	d	e	f	g	h

$\{a, b, c, d, e, f, g, h\} \times$

$\{1, 2, 3, 4, 5, 6, 7, 8\}$

$(e, 5) \in \{a, b, c, d, e, f, g, h\} \times$

$\{1, 2, 3, 4, 5, 6, 7, 8\}$

